MOHANRAJ K

AI/LLM Engineer | Agentic AI | RAG | LLM Evaluation | Backend Engineering

9629261512 · **moha.maraignaya@gmail.com** · Nagapattinam, Tamil Nadu · LinkedIn

SUMMARY

AI/LLM Engineer with **4+ years of experience** building and deploying production AI systems across **agentic AI, RAG, LLM evaluation, and AI automation**. Hands-on experience designing **LangGraph-based agent workflows**, developing and evaluating **end-to-end RAG pipelines**, and building **evaluation and observability workflows** for measuring **accuracy, retrieval quality, latency, cost, and reliability**. Strong backend engineering expertise with **Python, FastAPI, Django, asynchronous systems, and production AI APIs**.

EXPERIENCE

WORKHALL

Associate Software Engineer — AI/LLM Systems | Aug 2022 – Present

- Designed and implemented a **supervisor-based agent architecture using LangGraph**, orchestrating **6 specialized tools** through stateful workflows covering tool selection, multi-step orchestration, state, and memory management.
- Developed an end-to-end **DataFrame analysis agent** using LangChain's ReAct agent, enabling LLM-driven analysis across up to 5 dataframes with thousands of rows through dynamic dataframe/column selection and tool-based execution; reduced average token **usage by ~88%** by avoiding unnecessary dataset context.
- **Developed and evaluated end-to-end RAG pipelines** across document parsing, Azure-based heading extraction, embeddings, hierarchical/section-based chunking, and reranking; measured retrieval performance using **Recall@5, Recall@10, Precision, MRR, and NDCG**, identifying an Azure heading-based section chunking approach as the best-performing configuration.
- Benchmarked **15 LLM configurations** for document-to-form automation across **accuracy, automation, latency, hallucination, and cost**; identified **GPT-4.1** as the highest-accuracy configuration (**90.65%**) and **GPT-5.2** as the highest-automation configuration (**24.07%**) at **27.6s latency**.
- **Evaluated multiple speech-to-text models**, including OpenAI GPT-4o Transcribe, GPT-4o Mini Transcribe, Whisper, and Gemini, comparing **transcription accuracy, latency, and reliability**; built a real-time Whisper pipeline using a local-agreement algorithm, reducing **WER from 6.49 to 1.58 (76%)** while maintaining ~5-second latency.
- Built and deployed **30+ production AI APIs** using Python, FastAPI, and Django with **RabbitMQ-based asynchronous processing, Redis caching, MongoDB, and ClickHouse**.
- Fine-tuned **T5 Transformer models** for personalized welcome-message generation, contributing to dataset collection/curation and model evaluation to improve personalization and generation quality.

TECHNICAL SKILLS

Languages: Python
AI/LLM: Agentic AI, LLM Applications, RAG, Context Engineering, Tool Calling, Structured Outputs, LLM Evaluation, LLM Fine-tuning, OpenAI, Claude, Gemini, Llama
Frameworks: LangGraph, LangChain, LangSmith, FastAPI, Django
Backend & Data: REST APIs, RabbitMQ, Redis, MongoDB, ClickHouse, Asynchronous Processing, Docker
ML/Multimodal: PyTorch, Hugging Face Transformers, Whisper, OCR, pandas, scikit-learn

EDUCATION

- **PSG College of Technology** **B.E. — Electrical & Electronics Engineering**